mechanics, we'll think of this system as a qubit.

Volume I also discussed simple evolution laws that tell us how to update the state from instant to instant. The simplest law is just that nothing happens. In that case, if we go from one discrete instant (n) to the next ($n + 1$), the law of evolution is

$$\sigma(n+1) = \sigma(n). \qquad (1.1)$$

Let's expose a hidden assumption that we were careless about in *Volume I*. An experiment involves more than just a system to study. It also involves an apparatus $\mathcal{A}$ to make measurements and record the results of the measurements. In the case of the two-state system, the apparatus interacts with the system (the spin) and records the value of σ. Think of the apparatus as a black box[1] with a window that displays the result of a measurement. There is also a "this end up" arrow on the apparatus. The up-arrow is important because it shows how the apparatus is oriented in space, and its direction will affect the outcomes of our measurements. We begin by pointing it along the z axis (Fig. 1.1). Initially, we have no knowledge of whether $\sigma = +1$ or $\sigma = -1$. Our purpose is to do an experiment to find out the value of σ.

Before the apparatus interacts with the spin, the window is blank (labeled with a question mark in our diagrams). After it measures σ, the window shows a +1 or a −1. By looking at the apparatus, we determine the value of σ. That whole process constitutes a very simple experiment designed to measure σ.